Meat Science 97 (2014) 262–269

Contents lists available at ScienceDirect

Meat Science

journal homepage: www.elsevier.com/locate/meatsci

# Factors associated with offal, partial and whole carcass condemnation in ten French cattle slaughterhouses

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ARTICLE INFO

*Article history:*
Received 4 November 2013
Received in revised form 3 February 2014
Accepted 5 February 2014
Available online 15 February 2014

*Keywords:*
Cattle
Slaughterhouse
Condemnation
Syndromic surveillance

ABSTRACT

The proportion of cattle with offal, partial or whole carcass condemnation could be a useful indicator for animal health syndromic surveillance purposes. It requires first highlighting the factors associated with condemnation in order to include them in a modeling process. This study aims to identify and quantify these factors. It was based on data from ten French cattle slaughterhouses from 2006 to 2010 ($n = 1{,}439{,}868$ cattle). Multivariable multinomial logistic regression analyses were performed. Sex, age and slaughterhouse were the main effects for offal, partial and whole carcass condemnation and have to be taken into account when implementing syndromic surveillance systems based on meat inspection data. The presence of an error in cattle identification was identified as a potential indicator for a higher risk of condemnation and should be explored as a potential factor for risk-based meat inspection.



## 1. Introduction

The slaughterhouse, a focal point in the farm-to-fork chain for meat products, provides meat inspection data not otherwise available. European regulation requires an ante-mortem and post-mortem inspection of every animal slaughtered so as to guarantee that meat is fit for human consumption (European Parliament, 2004). Veterinary services are required to inspect for signs or lesions that could indicate a danger (e.g. acute peritonitis) or an organoleptic quality problem for meat consumption (e.g. chronic peritonitis) and remove any such products from the meat chain. Meat inspection data has to be properly registered.

Considering the food safety goal of meat inspection, these data are mainly pre-diagnostic data (except for regulated diseases such as tuberculosis) that could be used for syndromic surveillance purpose. Syndromic surveillance can be defined as the monitoring of non-specific health indicators including clinical signs, symptoms or proxy measures to enable the early identification of the impact (or absence of impact) of potential human or veterinary public health threats. It is implicit that the data are usually collected for purposes other than surveillance and, if possible, are automatically generated so as not to impose an additional burden on the data providers (Triple-S. Project, 2011). Meat inspection data could fit the definition of non-specific health indicator and then be used to implement a syndromic surveillance system for emerging disease detection (Dupuy et al., 2013b).

Cattle condemnations are commonly classified into three levels, i) whole carcass condemnations that mainly reflect a widespread infection, an acute stage of infection or any disorder that can have an impact on the whole carcass, ii) partial carcass condemnations that mainly reflect a chronic infection or a localized phenomenon, iii) offal condemnations that are mainly due to localized parasitic infection or any infection with a dedicated offal tropism. The proportion of cattle in each of these levels could be a useful indicator to monitor for syndromic surveillance purposes. The detection of abnormal changes in trend of the proportion of cattle in each of these three levels could reflect the occurrence of a known or unknown disease.

Until recently, the use of computerized methods to collect these data was difficult due to logistical issues (speed of the slaughter line, high moisture level and temperature variations incompatible with the use of IT equipment, together with the complexity of the data, due to the high diversity of possible reasons for condemnation of either whole carcasses or portions) (Dupuy et al., 2013c). Canada, Thailand, Finland, Denmark and Sweden have already implemented meat inspection databases (Alton et al., 2010, 2012; Kouvtanovitch et al., 2004; Ruoho et al., 2010; Tulayakul et al., 2008). In France, since 2005, a pilot system named *Nergal-Abattoir* was set up to collect data in real time during the slaughtering process. Ten cattle slaughterhouses accounting for about 20% of cattle slaughtered in the country were involved in this project.

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http://dx.doi.org/10.1016/j.meatsci.2014.02.008
0309-1740/

Meat inspection data are not yet used for syndromic surveillance (Dorea et al., 2011; Dupuy et al., 2013a). Initiatives are ongoing in Canada for cattle (Alton et al., 2010, 2012) and swine (Thomas-Bachli et al., 2012) and in the United States for swine (Weber, 2009). In order to use indicators based on meat inspection data for syndromic surveillance, it is necessary to identify factors that have an impact on such indicators. These factors will then have to be included in the modeling process so as to limit the number of alerts only due, particularly, to modification in slaughtered population characteristics. It also simplifies interpretation of potential alerts (Kleinman et al., 2005; Robertson et al., 2010). A first study was performed by Alton et al. on the factors associated with cattle whole carcass condemnation rate (Alton et al., 2010) and with portion condemnation rate (pneumonic lung and parasitic liver condemnation) (Alton et al., 2012) in provincially-inspected slaughterhouses in Ontario from 2001 to 2007.

Identifying factors associated with condemnation is the first step for the implementation of a syndromic surveillance system based on meat inspection data. It is also useful for farmers to understand which factors could influence condemnation so as to fully grasp their financial impact. Veterinary services are also interested in increasing knowledge on factors associated with condemnation to improve meat inspection efficiency. The decision taken by an official inspector when an abnormality is detected on a carcass is indeed the treatment of the carcass if possible (e.g. freezing for bovine cysticercosis) or the condemnation of all or part of the carcass. These measures have a direct financial impact for farmers due to the amount of meat condemned and reductions of the carcass price (Edwards et al., 1997; INTERBEV, 2007). Traditional meat inspection efficiency is increasingly discussed in a context of financial restriction. One possible means of improvement could be a risk-based meat inspection program, where animals identified at high-risk would receive an in-depth inspection whereas animals identified at low-risk would undergo a visual inspection only (Edwards et al., 1997). Factors associated with condemnation identified through this study could be used for this purpose as a way to identify which kind of animals could be considered at high or low-risk.

This study aimed to identify and quantify factors associated with offal, partial and whole carcass condemnation, based on data from ten French cattle slaughterhouses.

## 2. Material and methods

### 2.1. Data collection

The *Nergal-Abattoir* project used for the study was implemented from 2005 to 2010 by the French Ministry of Agriculture to collect data in real time during the slaughtering process in ten cattle slaughterhouses. Cattle more than eight months old slaughtered in these ten slaughterhouses from January 2006 to December 2010 were included in the study. Veal calves (cattle under eight months old) were excluded because the farming practices, the commercial network of this animal category as well as the typology of lesions observed during meat inspection (Dupuy et al., 2013c) were all very specific to this age category. Cattle euthanized or that died in the ante-mortem inspection area were also excluded.

Data were collected using touch screens on the slaughter lines, provided by the food business operator. Data were then transmitted through a constant data flow to the database of the French Ministry of Agriculture. The main objectives of this system were to guarantee the traceability of meat inspection results and to automatically produce the mandatory condemnation reports for cattle owners (Dupuy et al., 2013c). For each animal slaughtered, the *Nergal-Abattoir* database contained: animal identification number (ID), slaughterhouse ID, dates of birth and slaughter, last farm location, sex, breed, information on the presence or absence of any error in cattle identification, absence or presence of condemnation and, in the latter case, reasons for condemnation, and condemnation locations.

The French National Cattle Register (BDNI) (Perrin et al., 2012) was used to collect information regarding animal movements from birth to slaughterhouse. From these data mandatorily requested from farmers, it was possible to define the number of different farms in which each animal was raised and breed from birth to slaughterhouse.

### 2.2. Data analysis

We subjected the data to multivariable multinomial logistic regression analysis also called polytomous logistic regression. Based on meat inspection results, the outcome nominal variable (Y) was defined as the absence of condemnation ($Y = 0$), the condemnation of offal(s) only ($Y = 1$), the condemnation of part of the carcass whether or not it included offal condemnation ($Y = 2$) and condemnation of the whole carcass ($Y = 3$). Each animal could only be attributed to a single category. Nine variables (detailed below) were examined for their association with the outcome variable. Sex, age, production type, slaughterhouse ID, month and year of slaughter and the distance between the last farm location and the slaughterhouse were evaluated because their relevancy was already showed (Alton et al., 2010). The presence of an error in cattle identification was evaluated because it could be considered as a proxy measure of the quality level of farm management. The number of farms where the animal had lived was investigated because we hypothesized that the higher the number of farms the higher the risk to have been infected by a disease and thus to present lesions at slaughterhouse.

Missing data were considered as missing at random regarding the outcome variable. It was thus decided to exclude cattle with missing data and not impute missing data values (Donders et al., 2006).

Cattle breeds were grouped according to production type as defined by the French national organization of agriculture products, into the categories “dairy”, “beef” and “mixed” cattle (FranceAgriMer, 2011). The ages of cattle were classified according to European Regulation (European Parliament, 2007) and farming practices into four different age groups: (8, 24) months, (2, 3.5) years, (3.5, 5) years, and ≥5 years. Slaughterhouses were coded (1 to 10) to preserve anonymity. The presence of an error in cattle identification was used as a binary variable: presence/absence. The number of farms where each animal had lived was classified into two levels, “one farm” and “more than one farm”. The distance between the last farm location of the animal and the slaughterhouse was calculated in kilometers using distance as the crow flies between the *commune* of the farm and the slaughterhouse. *Commune* is the smallest administrative area in France. GPS coordinates of the center of each *commune* were used. This variable was tested not only as a numeric variable but also as a categorical variable using quartiles as breakpoints. Correlations between covariables were addressed using the Cramer's V measure of association. If any pair of variables was found to be strongly correlated, i.e. higher than 0.8, then the two correlated variables were replaced by a combined variable. For instance combining one variable with i categories to one variable with j categories consists in merging each of the i categories of the first variable with each of the j categories of the second variable. The combined variable has then $i * j$ categories.

The type of multinomial logistic model used was generalized logit model. The modeling selection strategy of Hosmer and Lemeshow was used (Hosmer and Lemeshow, 2000). In a first step, each variable was evaluated separately for statistical significance. Any variable with a p-value lower than 0.05 at this univariate step was included in a multivariable model and a backward step-wise approach was used to select the final model. We then evaluated the potential confounding effect of variables by a backward step-by-step procedure, checking if the removal of non-significant variables produced important changes in the coefficient of the other variables in the model. There was no need to use a more conservative level (e.g. $p < 0.2$ as recommended by Hosmer and Lemeshow) for variable selection at the univariate step because of the high power of analysis due to the large population size. Interactions that make sense from a biological perspective were tested, i.e. interactions between sex, age and production type.

A random effect for the last farm location was also evaluated using three separate models (Y = 1/Y = 0; Y = 2/Y = 0 and Y = 3/Y = 0) due to technical issues (large population size). The extent of the random effect was evaluated by comparing the odds ratios of fixed effects in the models with and without random effect (overlapping or not of confidence intervals).

For the comparison of nested models, likelihood ratio tests were used, otherwise the Akaïke information criterion (AIC) was used. As recommended by Hosmer and Lemeshow, we assessed the goodness of fit using the individual logistic regressions approach of Begg and Gray (Hosmer and Lemeshow, 2000). Then, for each individual logistic regression, we used area under the receiver operating curve (ROC curve) for assessment of fit. It provides a measure of the model's ability to discriminate between cattle with or without condemnation (offal, partial or whole carcass). The closer to unity the value was the higher the ability of the model to discriminate.

Calculations were performed with R software's “nnet”, “verification”, “vcd”, “effects” and “lme4” packages (R Development Core Team, 2010).

## 3. Results

From January 2006 to December 2010, 1,629,080 cattle more than eight months old passed through the ten slaughterhouses involved in the *Nergal-Abattoir* project. Cattle euthanized (n = 1029) or that died in the ante-mortem inspection area (n = 290), were excluded from this study. Cattle with missing data (n = 187,893) were also excluded. The study population involved 1,439,868 cattle more than eight months old.

The number of days with available data for each slaughterhouse varied from 345 to 1694 because the *Nergal-Abattoir* project did not start and finish at the same time for each slaughterhouse (Table 1). The description of the study population is presented in Table 2. The main motives for offal condemnation were abscess (19.2%), liver fluke (18.9%) and sclerosis of the liver (9.9%). For partial condemnation with or without offal condemnation, the main motives were abscess (18.4%), infiltration (15.7%) and arthritis (13.5%). Spoilage, for instance abnormal maturation of the meat (e.g. Dark Firm Dry meat) (37.1%) and peritonitis (18.1%) were the main reasons for whole carcass condemnation.

Correlation between all pairs of variables was lower than 0.8 (ranging from 0.001 and 0.4) except between age and sex (0.83). A combined variable Sex–Age was thus created. All variables used for univariable statistical analysis were significantly associated with the outcome variable and were thus retained in the multivariable multinomial logistic regression. As sex and age were combined, only one interaction is needed to be evaluated, i.e. between Sex–Age and production type, and it was statistically significant. This interaction was included in the model. Areas under the ROC curve were 0.68, 0.65 and 0.73 for the models Y = 1/Y = 0, Y = 2/Y = 0 and Y = 3/Y = 0 respectively. The odds ratios of the models with and without the random effect were similar, which is why we did not retain the mixed effect model.

### 3.1. Sex effect

For dairy and mixed cattle, the odds of condemnation for females were higher than for males except for cattle more than five years old and for dairy cattle in the case of offal condemnation (Fig. 1 and Table 3). For beef cattle, the odds of condemnation for males were higher than females except for cattle from 8 to 24 months old (Fig. 1 and Table 3).

### 3.2. Production type effect

For cattle from 2 to 5 years old, the odds of condemnation were highest for dairy cattle and lowest for beef cattle, with mixed cattle falling between the two. For males, concerning partial and whole carcass condemnation, the differences regarding production type were not significant (Fig. 2 and Table 3). Females over five years old presented the same profile for offal condemnation. Production type effect was limited for cattle from 8 to 24 months old and over five years old except for females for offal condemnation as previously mentioned and for females for whole carcass condemnation, for which the odds concerning beef cattle were lower than for mixed and dairy cattle (Fig. 2 and Table 3).

### 3.3. Age effect

The global profile of age effect revealed increasing odds with age whatever the type of condemnation. The age effect was low for females for whole carcass condemnation and for all cattle, irrespective of the type of condemnation, from two to five years old (Figs. 1 and 2). For whole and partial carcass condemnation, age effect was higher for males than for females whatever the production type (Fig. 1).

### 3.4. Other effects

Depending on the slaughterhouse, odds ratios for offal condemnation ranged from 1 to 4.97, for partial condemnation from 1 to 4.83 and for whole condemnation from 1 to 4.43 (results obtained after a reference switch) (Table 4). Each slaughterhouse presented a different profile regarding condemnation, none having the lowest odds or the highest odds for all types of condemnation. The effects of month of slaughter, year of slaughter, distance between farm and slaughterhouse, number of farms where animal had lived and the presence or not of an

**Table 1**
Description of data availability in the *Nergal-Abattoir* project database in the ten slaughterhouses involved.

| Slaughterhouse | First day of data availability | Last day of data availability | Number of days of data availability | Mean number of cattle slaughtered each day | Number of cattle slaughtered (%) |
|---|---|---|---|---|---|
| 1 | 12/05/2006 | 31/12/2010 | 1694 | 270 | 281,262 (19.53) |
| 2 | 20/06/2006 | 25/03/2010 | 1374 | 241 | 225,553 (15.66) |
| 3 | 23/11/2006 | 22/12/2010 | 1490 | 84 | 74,084 (5.15) |
| 4 | 12/10/2006 | 11/05/2009 | 942 | 216 | 137,321 (9.54) |
| 5 | 08/10/2007 | 17/09/2008 | 345 | 129 | 28,696 (1.99) |
| 6 | 02/01/2006 | 09/10/2009 | 1376 | 352 | 334,904 (23.26) |
| 7 | 06/03/2007 | 17/09/2010 | 1291 | 101 | 65,956 (4.58) |
| 8 | 27/03/2007 | 18/09/2008 | 541 | 135 | 49,642 (3.45) |
| 9 | 26/06/2007 | 19/01/2010 | 938 | 232 | 150,071 (10.42) |
| 10 | 15/03/2006 | 01/02/2008 | 688 | 195 | 92,379 (6.42) |
| Total | | | | | 1,439,868 (100) |

**Table 2**
Description of cattle more than 8 months old slaughtered in one of the ten French *Nergal-Abattoir* slaughterhouses from 2006 to 2010.

| | Proportion (%) of cattle with | | | | Total (number of animals) |
|---|---|---|---|---|---|
| | Absence of condemnation | Only offal condemnation | Partial carcass condemnation | Whole carcass condemnation | |
| Sex–Age | | | | | |
| Male (8, 24) months | 86.9 | 11.1 | 1.8 | 0.2 | 458,455 |
| Female (8, 24) months | 83.8 | 12.9 | 2.7 | 0.7 | 9992 |
| Male (2, 3.5) years | 84.7 | 13.3 | 1.7 | 0.3 | 105,104 |
| Female (2, 3.5) years | 83.9 | 13.5 | 2.0 | 0.6 | 167,260 |
| Male (3.5, 5) years | 78.5 | 17.9 | 3.0 | 0.6 | 18,560 |
| Female (3.5, 5) years | 77.8 | 18.3 | 2.9 | 1.0 | 180,937 |
| Male ≥5 years | 64.2 | 28.4 | 6.4 | 1.1 | 17,028 |
| Female ≥5 years | 67.8 | 27.1 | 4.0 | 1.1 | 482,532 |
| Production type | | | | | |
| Beef | 80.2 | 17.0 | 2.5 | 0.4 | 702,900 |
| Dairy | 75.9 | 20.1 | 3.1 | 1.0 | 452,356 |
| Mixed | 78.3 | 17.7 | 3.0 | 1.0 | 284,612 |
| Month of slaughter | | | | | |
| January | 78.2 | 18.2 | 3.0 | 0.6 | 125,456 |
| February | 79.0 | 17.4 | 3.0 | 0.6 | 105,451 |
| March | 79.9 | 16.8 | 2.8 | 0.6 | 109,284 |
| April | 79.4 | 17.2 | 2.8 | 0.7 | 109,585 |
| May | 80.3 | 16.5 | 2.6 | 0.6 | 109,273 |
| June | 79.7 | 17.0 | 2.7 | 0.7 | 118,746 |
| July | 78.3 | 18.5 | 2.5 | 0.7 | 129,328 |
| August | 77.4 | 19.3 | 2.6 | 0.7 | 134,730 |
| September | 77.6 | 18.7 | 2.9 | 0.8 | 127,731 |
| October | 76.8 | 19.7 | 2.8 | 0.7 | 129,760 |
| November | 77.2 | 19.4 | 2.8 | 0.7 | 120,962 |
| December | 78.5 | 17.9 | 2.9 | 0.7 | 119,562 |
| Year of slaughter | | | | | |
| 2006 | 79.6 | 17.4 | 2.4 | 0.6 | 216,494 |
| 2007 | 79.2 | 17.6 | 2.7 | 0.6 | 424,555 |
| 2008 | 78.9 | 17.7 | 2.7 | 0.7 | 434,360 |
| 2009 | 76.4 | 19.6 | 3.2 | 0.8 | 335,069 |
| 2010 | 75.7 | 20.3 | 3.4 | 0.6 | 29,390 |
| Slaughterhouse | | | | | |
| 1 | 84.1 | 13.5 | 1.9 | 0.5 | 281,262 |
| 2 | 70.2 | 25.4 | 3.8 | 0.6 | 225,553 |
| 3 | 71.3 | 25.6 | 1.9 | 1.2 | 74,084 |
| 4 | 90.2 | 7.6 | 1.5 | 0.8 | 137,321 |
| 5 | 88.7 | 9.0 | 1.3 | 1.0 | 28,696 |
| 6 | 81.1 | 14.2 | 3.7 | 0.9 | 334,904 |
| 7 | 77.2 | 20.5 | 1.7 | 0.6 | 65,956 |
| 8 | 64.6 | 30.7 | 4.2 | 0.5 | 49,642 |
| 9 | 69.8 | 26.7 | 3.0 | 0.6 | 150,071 |
| 10 | 79.3 | 18.4 | 2.0 | 0.2 | 92,379 |
| Identification anomaly | | | | | |
| Absence | 78.5 | 18.1 | 2.8 | 0.7 | 1,435,499 |
| Presence | 66.1 | 26.9 | 5.8 | 1.3 | 4369 |
| Number of farms | | | | | |
| =1 | 78.7 | 17.9 | 2.8 | 0.7 | 991,550 |
| >1 | 78.0 | 18.6 | 2.7 | 0.7 | 448,318 |
| Farm-slaughterhouse distance | | | | | |
| 0−33 | 80.0 | 16.8 | 2.6 | 0.6 | 358,728 |
| 34−64 | 79.4 | 17.3 | 2.8 | 0.6 | 353,875 |
| 65−129 | 79.4 | 17.3 | 2.8 | 0.6 | 364,189 |
| >=130 | 75.1 | 21.0 | 3.0 | 0.9 | 363,076 |
| Total | 78.5 | 18.1 | 2.8 | 0.7 | 1,439,868 |

error in cattle identification were significant but low. The month of slaughter effect was higher for whole carcass condemnation than for partial or offal condemnation. Year of slaughter effect was higher for partial condemnation than for whole condemnation or offal condemnation. The distance between farm and slaughterhouse effect as well as the number of farms effect were higher for whole condemnation than for partial or offal condemnation. The presence of an error in cattle identification was a factor significantly associated with condemnation, whatever the type.

## 4. Discussion

The objective of this study was to identify factors associated with offal, partial and whole carcass condemnation. The main factors identified were age, sex and slaughterhouse. The method used and the interpretation of identified factors are discussed in this section.

### 4.1. Data analysis

The ten slaughterhouses included in the *Nergal-Abattoir* project were all part of the same food chain operator. There was no habit known to be specific to these ten slaughterhouses compared to others in France. The main factors that could differ from one slaughterhouse to another were the characteristics of the slaughtered population. This has been taken into account in our study by including sex, age and production type in the modeling process. We could not quantify the bias induced by the use of these ten slaughterhouses but we could hypothesize that it was limited.

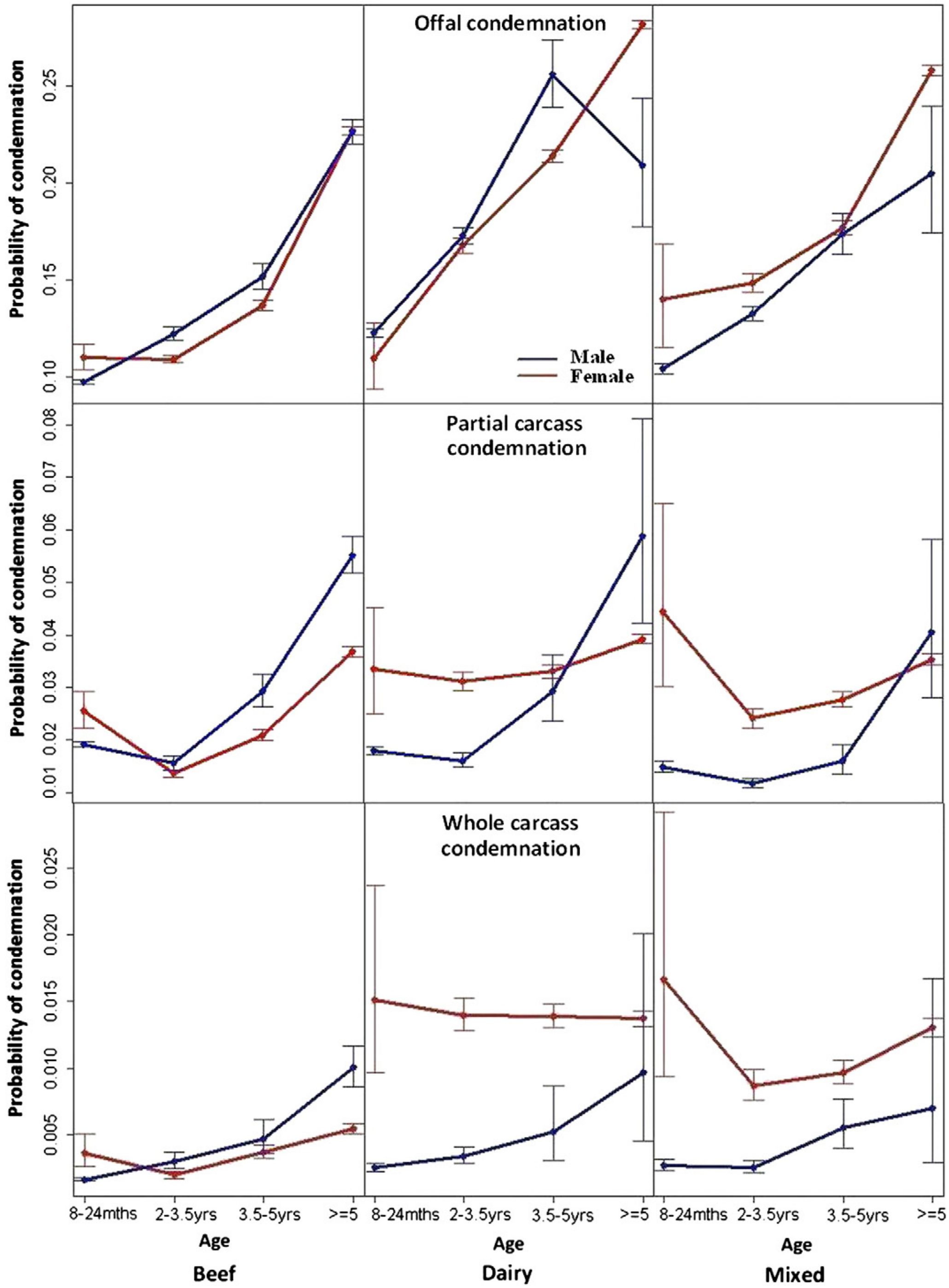


**Fig. 1.** Probability of offal condemnation, partial carcass condemnation and whole carcass condemnation according to sex, age and production type.

A random effect on the last farm location variable was investigated based on the hypothesis that animals from the same herd were more likely to have a similar lesion profile than animals from different herds. This hypothesis was certainly more relevant for whole carcass condemnation (acute lesions) than for partial or offal condemnation (more chronic lesions). Taking into account a random effect for last farm location involves the hypothesis that the origin of lesions detected through meat inspection could be attributed to the last farm, which is much more credible for acute lesions (whole carcass condemnation) than for more chronic lesions (partial or offal condemnation). In any case, we did not retain models with random effect on last farm location even if the effect was statistically significant because the odds ratios of the models with and without this effect were similar.

Areas under the ROC curve were large enough to ensure a good ability of the models to discriminate between cattle with or without condemnation (offal, partial or whole).

### 4.2. Interpretation of factors associated with condemnation

Sex, age and slaughterhouse were the main effects for offal, partial and whole carcass condemnation.

The slaughterhouse effect is not easy to interpret. Indeed, slaughterhouse and the location of the last farm are known to be correlated.

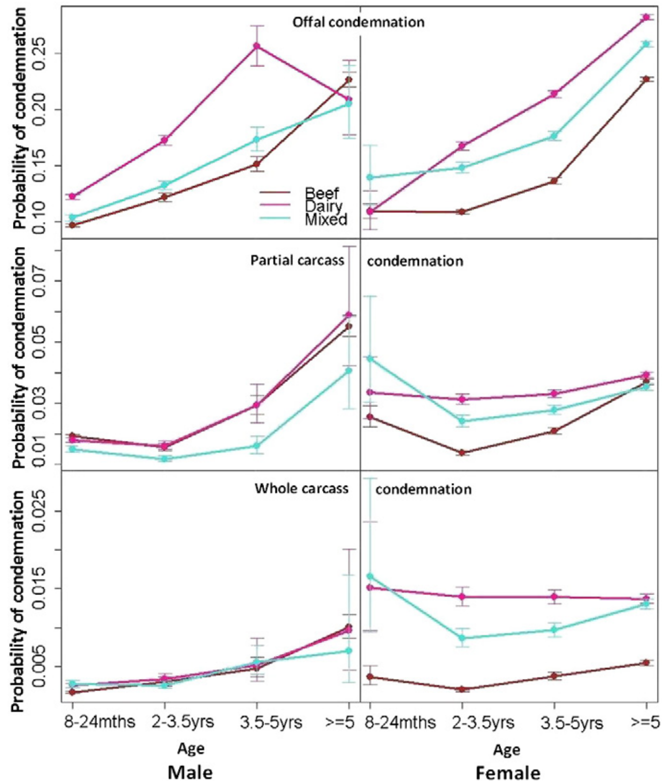


**Fig. 2.** Probability of offal condemnation, partial carcass condemnation and whole carcass condemnation according to production type, age and sex.

Consequently, the slaughterhouse effect presented in Table 2 certainly reflects the combination of 1) a real slaughterhouse effect (size of slaughterhouse, meat inspection line speed, level of experience of the official inspectors, specialization of certain slaughterhouses in a dedicated category of cattle, etc.), 2) a software practice effect (differences in the use of the software for recording condemnations even if the database was identical in each slaughterhouse), and 3) a region effect (geographical variation of disease occurrence).

**Table 3**
Estimates from the final multivariable multinomial logistic regression model of the odds of offal condemnation (Y = 1), partial condemnation (Y = 2) and whole carcass condemnation (Y = 3). Odds ratios and 95% Confidence Interval (95% CI) are presented for each combination of production type, sex and age.

| | Production type | | |
|---|---|---|---|
| | Beef | Dairy | Mixed |
| *Y = 1/Y = 0* | | | |
| Male (8, 24) months | 1 | 1.30 [1.27;1.33] | 1.08 [1.04;1.11] |
| Female (8, 24) months | 1.16 [1.08;1.24] | 1.18 [0.99;1.41] | 1.59 [1.27;1.99] |
| Male (2, 3.5) years | 1.29 [1.25;1.34] | 1.94 [1.83;2.06] | 1.41 [1.33;1.5] |
| Female (2, 3.5) years | 1.13 [1.1;1.16] | 1.94 [1.86;2.02] | 1.65 [1.57;1.73] |
| Male (3.5, 5) years | 1.69 [1.6;1.78] | 3.28 [2.92;3.7] | 1.96 [1.76;2.17] |
| Female (3.5, 5) years | 1.48 [1.45;1.52] | 2.63 [2.53;2.73] | 2.04 [1.96;2.13] |
| Male ≥5 years | 2.91 [2.8;3.02] | 2.62 [2.13;3.24] | 2.49 [2.02;3.07] |
| Female ≥5 years | 2.82 [2.77;2.87] | 3.85 [3.76;3.94] | 3.38 [3.3;3.47] |
| *Y = 2/Y = 0* | | | |
| Male (8, 24) months | 1 | 0.96 [0.91;1.01] | 0.78 [0.72;0.84] |
| Female (8, 24) months | 1.36 [1.18;1.57] | 1.83 [1.34;2.5] | 2.55 [1.7;3.82] |
| Male (2, 3.5) years | 0.83 [0.76;0.91] | 0.91 [0.79;1.06] | 0.63 [0.55;0.73] |
| Female (2, 3.5) years | 0.71 [0.67;0.76] | 1.82 [1.65;2.01] | 1.35 [1.21;1.51] |
| Male (3.5, 5) years | 1.65 [1.48;1.85] | 1.89 [1.44;2.49] | 0.92 [0.72;1.16] |
| Female (3.5, 5) years | 1.15 [1.08;1.22] | 2.06 [1.88;2.25] | 1.62 [1.47;1.79] |
| Male ≥5 years | 3.58 [3.33;3.86] | 3.74 [2.6;5.38] | 2.50 [1.68;3.71] |
| Female ≥5 years | 2.32 [2.23;2.41] | 2.71 [2.57;2.85] | 2.34 [2.21;2.47] |
| *Y = 3/Y = 0* | | | |
| Male (8, 24) months | 1 | 1.55 [1.35;1.78] | 1.61 [1.35;1.93] |
| Female (8, 24) months | 2.24 [1.59;3.14] | 9.33 [5.87;14.81] | 10.89 [6.09;19.47] |
| Male (2, 3.5) years | 1.84 [1.48;2.28] | 2.20 [1.56;3.11] | 1.59 [1.12;2.25] |
| Female (2, 3.5) years | 1.20 [1.02;1.4] | 9.20 [7.37;11.49] | 5.49 [4.3;7] |
| Male (3.5, 5) years | 3.01 [2.27;4] | 3.82 [2.01;7.25] | 3.59 [2.16;5.96] |
| Female (3.5, 5) years | 2.32 [1.99;2.71] | 9.75 [7.91;12.03] | 6.38 [5.12;7.96] |
| Male ≥5 years | 7.36 [6.17;8.77] | 6.88 [3.15;15.03] | 4.85 [1.96;12.02] |
| Female ≥5 years | 3.89 [3.48;4.34] | 10.67 [9.36;12.16] | 9.73 [8.5;11.13] |

The global tendency for age effect was that the older the animal, the higher the probability that it would have suffered a lesion at some time in its life.

The extent of the sex effect for whole carcass condemnation for dairy and mixed cattle (females had higher odds than males) could be explained by a major difference in farming practices according to sex: females are raised outside whereas males are raised inside.

The month of slaughter effect was greater for whole carcass condemnation than for partial or offal condemnation. We could hypothesize a dilution of the effect for chronic conditions (partial or offal condemnation) due to the diversity of the delay between the occurrence of the disease and the observation of the lesion in the slaughterhouse. The month effect on whole carcass condemnation in this study (odds ratio from 1 to 1.52) was similar to the season effect (from 1 to 1.32) in another study in Ontario (Alton et al., 2010) but in France the lowest probability of whole carcass condemnation was observed in the winter season (from January to March) whereas in Ontario winter was the season of highest risk. The higher probability of whole carcass condemnation in spring in France could be explained by the calving season being responsible for obstetric disorders. The higher probability observed in summer could be explained by the increasing incidence of traumatic reticulo-peritonitis during this pasture period. Furthermore, in winter, most cattle are raised inside, meaning that farmers can better observe them and rapidly treat them which could explain the lower probability of condemnation during this season.

Our results cannot be directly compared to those of Alton et al. for age, sex and production type effect because only one categorical variable with four levels (“calves”, “cows”, “heifers”, “steers”) was used (Alton et al., 2010, 2012). We were able to confirm, however, that it is important to take information regarding animal characteristics (sex,

**Table 4**
Estimates from the final multivariable multinomial logistic regression model of the odds of offal condemnation (Y = 1), partial condemnation (Y = 2) and whole carcass condemnation (Y = 3). Odds ratios and 95% Confidence Interval (95% CI) are presented for month and year of slaughter, slaughterhouse ID, presence of an error in cattle identification, distance between the last farm location and the slaughterhouse and the number of farms where the animal had lived.

| | Outcome variable | | |
|---|---|---|---|
| | Y = 1/Y = 0 | Y = 2/Y = 0 | Y = 3/Y = 0 |
| *Month of slaughter* | | | |
| January | 1 | 1 | 1 |
| February | 0.97 [0.95;0.99] | 1.02 [0.97;1.07] | 0.94 [0.85;1.05] |
| March | 0.95 [0.93;0.97] | 0.98 [0.93;1.03] | 0.96 [0.87;1.07] |
| April | 0.97 [0.95;0.99] | 0.99 [0.94;1.04] | 1.22 [1.1;1.35] |
| May | 0.93 [0.91;0.95] | 0.94 [0.9;0.99] | 1.22 [1.1;1.36] |
| June | 0.95 [0.93;0.97] | 1 [0.95;1.05] | 1.33 [1.2;1.47] |
| July | 1.02 [0.99;1.04] | 0.96 [0.91;1.01] | 1.49 [1.35;1.64] |
| August | 1.07 [1.04;1.09] | 1.01 [0.96;1.06] | 1.37 [1.24;1.52] |
| September | 1.02 [1;1.04] | 1.10 [1.05;1.16] | 1.52 [1.38;1.68] |
| October | 1.07 [1.04;1.09] | 1.08 [1.03;1.14] | 1.29 [1.16;1.42] |
| November | 1.04 [1.01;1.06] | 1.10 [1.04;1.15] | 1.16 [1.05;1.29] |
| December | 0.94 [0.92;0.97] | 1.12 [1.07;1.18] | 1.21 [1.09;1.34] |
| *Year of slaughter* | | | |
| 2006 | 1 | 1 | 1 |
| 2007 | 0.93 [0.91;0.94] | 1.23 [1.19;1.28] | 1.12 [1.04;1.2] |
| 2008 | 0.89 [0.88;0.91] | 1.25 [1.21;1.3] | 1.2 [1.12;1.29] |
| 2009 | 0.94 [0.93;0.95] | 1.55 [1.49;1.6] | 1.32 [1.23;1.42] |
| 2010 | 0.92 [0.88;0.95] | 1.87 [1.73;2.02] | 1.53 [1.29;1.82] |
| *Slaughterhouse* | | | |
| 1 | 1.67 [1.6;1.75] | 1.50 [1.35;1.67] | 0.56 [0.49;0.63] |
| 2 | 3.47 [3.32;3.62] | 3.19 [2.87;3.55] | 0.90 [0.79;1.03] |
| 3 | 3.45 [3.3;3.62] | 1.61 [1.43;1.81] | 1.39 [1.21;1.61] |
| 4 | 1.04 [0.99;1.09] | 1.33 [1.19;1.49] | 1.20 [1.05;1.37] |
| 5 | 1 | 1 | 1 |
| 6 | 1.67 [1.6;1.75] | 3.29 [2.96;3.65] | 0.97 [0.86;1.1] |
| 7 | 4.19 [4;4.39] | 1.79 [1.59;2.03] | 1.42 [1.21;1.67] |
| 8 | 4.97 [4.75;5.21] | 4.83 [4.32;5.41] | 0.71 [0.6;0.85] |
| 9 | 4.47 [4.28;4.67] | 2.77 [2.49;3.09] | 0.91 [0.79;1.04] |
| 10 | 2.37 [2.26;2.48] | 1.95 [1.74;2.19] | 0.32 [0.26;0.38] |
| *Cattle identification error* | | | |
| Absence | 1 | 1 | 1 |
| Presence | 1.46 [1.36;1.56] | 1.89 [1.66;2.16] | 1.94 [1.49;2.52] |
| *Number of farms* | | | |
| =1 | 1 | 1 | 1 |
| >1 | 1.12 [1.11;1.14] | 1.11 [1.08;1.13] | 1.28 [1.23;1.34] |
| *Farm-slaughterhouse distance* | | | |
| [0–34] | 1 | 1 | 1 |
| [34–65] | 1.06 [1.05;1.07] | 0.98 [0.95;1.01] | 0.92 [0.87;0.98] |
| [65–130] | 1.01 [1;1.02] | 0.94 [0.91;0.97] | 0.92 [0.87;0.98] |
| ≥130 | 1.06 [1.05;1.07] | 0.97 [0.94;1] | 1.23 [1.16;1.3] |

age, production type) into account when adjusting a model for syndromic surveillance based on meat inspection data. Some animals are indeed more likely to be processed in certain slaughterhouses because they are specialized in a dedicated animal class (e.g. old dairy cattle). It could also be relevant to take into account the carcass quality notation as suggested by the studies of Alton et al. (2010, 2012) because certain slaughterhouses accept a larger proportion of poorer quality cattle due to dedicated commercial channels. This information which could only be provided by the food business operator, was not available for this study.

The small effect of the distance between the farm and the slaughterhouse for partial condemnation was surprising; we had supposed that the longer the transport process the higher the condemnation rate, because of the possible occurrence of bruising (Strappini et al., 2009). It could reflect an improvement regarding transportation conditions linked to the application of European Regulations on animal transportation. It could also be explained by a common practice in some slaughterhouses of trimming off bruises from carcasses directly on the slaughter line without registering the information instead of performing an official partial condemnation. Finally, it can be due to the fact that, for most of the cattle (99%), the transportation distance was lower than 350 km which was perhaps too low to exert a visible negative effect on partial condemnation.

The odds of having a condemnation (offal, partial and whole carcass) increased if the animals had lived in more than one farm. This could be explained by the increase of infection burden linked to the multiplication of farms and thus the occurrence of possible lesions that could be detected during meat inspection.

Having an error in cattle identification was identified as a factor associated with condemnation irrespective of the type of condemnation. The presence of an error in cattle identification can be considered as a proxy measure of the quality level of farm management. We could hypothesize that a farmer facing difficulties in identifying his animals will probably face other management issues that could have an impact on animal health and thus on lesions detected during the meat inspection process.

Meat price information for each cattle type was not available. Alton et al. demonstrated an impact of meat price on whole carcass condemnation but not on partial carcass condemnation (Alton et al., 2010, 2012). It seems obvious that meat price has an impact on the quality and the number of animals being shipped to slaughter as it is part of the decision-making process by farmers in when to cull their animals. This factor would have to be taken into account for the interpretation of alert indicators in a syndromic surveillance system based on meat inspection data, especially if the indicator is the number of cattle condemned and not a proportion of cattle condemned. Other factors such as raw food price or environmental events that could have an impact on grazing (e.g. drought) also have to be taken into account considering their possible impact on carcass weight and more generally on animal health.

### *4.3. Interest for syndromic surveillance*

The factors identified to have the major effect on offal, partial and whole carcass condemnation were age, sex and slaughterhouse. These variables will have to be taken into account when using methods to detect abnormal trends or outliers in the proportion of cattle condemned in order to limit the number of false alarms. For instance, an increase in the condemnation rate could simply be due to an increase in the proportion of old cattle slaughtered as the proportion of condemnation increases with age. Including age in the modeling process will allow distinguishing an increase only due to age factor from an increase due to the occurrence of a disease.

## 5. Conclusions

This study highlighted that sex, age, production type, month and year of slaughter, slaughterhouse, presence of an error in cattle identification, distance between the last farm location and the slaughterhouse and the number of farms where the animal had lived all had an impact on cattle carcass condemnation. This study showed that these factors associated with condemnation, even if significant for each type of condemnation (offal, partial, whole), did not have the same importance for each type of condemnation. The major effects were linked to sex, age and slaughterhouse. These factors will have to be taken into account when implementing a syndromic surveillance system based on meat inspection data.

The presence of an error in cattle identification was identified as a potential indicator for a higher risk of condemnation and should be explored in the perspective of risk assessment-based meat inspection. More careful inspection of animals presenting an error of identification could easily be implemented considering the easy availability of the data and the relatively low number of animals concerned.

## Acknowledgments

The authors would like to acknowledge the French Ministry of Agriculture for funding this study and for having provided access to the data. They would like to thank Coup de Puce Expansion for checking the English.

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